SURP 2026: EndTerm Report

Note: We are initiating the end-term report submissions to gauge the level of development your projects have gone through from the start.

Instructions:

- Make a copy of this Google Document, complete the necessary fields in the provided form, and have it verified by the professor, including a digital signature.
- Download this document as a **PDF file** (mandatory) and submit it via the designated Google form: Endterm Submission Form
- Deadline: **11.59 PM, 4th September, 2026**
- All students pursuing the SURP Project need to submit the Endterm report duly signed by the professor for any future incentives/resume verification/ certificates.
- In case of any queries, kindly reach out to the EnPoWER Team.

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TO BE FILLED BY THE STUDENT

Details of the student

Name	**Kartikya Saini**
Roll Number	**24b4519**
Year of Study	**3rd year**
Contact number	[redacted]

Details of the project

Title of project	Quantitative Research & Trading Framework (Team A — Strategy Design)
Name of the professor's guide	Prof. Sandip Mondal (Dept. of Electrical Engineering, IIT Bombay)

Progress Update

Project status: On-going	**Would you like to continue the project after SURP? : Yes**

List the targets achieved in the project so far

- Implemented a market-neutral PCA-residual statistical-arbitrage strategy (Avellaneda–Lee framework) end-to-end, PCA factor decomposition, residual extraction, Ornstein-Uhlenbeck mean-reversion fit and s-score signal generation.
- Built a look-ahead-safe backtesting harness from scratch (signal → decision → execution → P&L) with triple-barrier exits, inverse-volatility sizing, dollar-neutral construction and a full Indian transaction-cost model.
- Ran the cash-equity study on 10 years of NSE data (2015–2024, 2,466 trading days, 435 stocks): +32.9% zero-cost vs −32.9% full-cost return, isolating an edge that is entirely erased by transaction costs.
- Identified the decisive structural finding via a cost-sensitivity sweep: break-even sits at ~18–20 bps round-trip, below the ~22 bps Securities Transaction Tax floor, so the strategy is untradable in cash equities under any execution assumption.
- Extended the identical strategy to NSE single-stock futures on a genuinely point-in-time F&O universe built from the free NSE bhavcopy archive continuous front-month contracts, corporate-action adjustment, whole-lot sizing, monthly expiry rolls and a realistic futures cost model.

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- Confirmed the cost hypothesis on futures (−32.9% under cash-delivery costs → +11.8% under futures costs), but found the return statistically indistinguishable from zero (Sharpe 0.28 ± 0.33) and the reversion signal insignificant out-of-sample.
- Re-ran the out-of-sample test on a point-in-time universe and established that the cash study's own multi-horizon significance was carried by survivorship bias, a self-correction of the earlier result.
- Validated the engine throughout: known-answer synthetic tests, independent equity-curve reconstruction matching to floating-point precision, look-ahead assertions across every fill, and an adversarial audit that caught and fixed a corporate-action accounting bug before any result was reported.

A summary of the work you have done till now in 7-8 lines

This project builds and rigorously evaluates a PCA-residual statistical-arbitrage strategy for Indian markets. The strategy uses Principal Component Analysis to strip common market and sector factors from stock returns, then trades the mean-reversion of the residual component in a market-neutral, dollar-neutral long/short book with triple-barrier exits and inverse-volatility sizing.

I implemented the full signal pipeline and a look-ahead-safe backtesting harness in Python and tested it on ten years of real NSE cash-equity data under a realistic Indian cost model. The result is a rigorous negative finding: the strategy returns +32.9% before costs and −32.9% after, because break-even (~18–20 bps round-trip) sits below the unavoidable Securities Transaction Tax floor (~22 bps). The edge is real in the data but structurally untradable in cash equities.

Because the cost analysis pointed precisely to a lower-cost instrument, I extended the identical strategy to NSE single-stock futures, building the entire data pipeline from the free NSE bhavcopy archive: a point-in-time F&O universe, continuous front-month contracts with expiry rolls, corporate-action adjustment, and whole-lot execution on real contracts. The cost

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hypothesis held, the same strategy turns from −32.9% to +11.8% — but the return is statistically indistinguishable from zero, concentrated in a single year, and does not survive a modest increase in the slippage assumption.

Critically, re-running the out-of-sample test on a point-in-time universe showed that the cash study's own statistical significance had been carried largely by survivorship bias, and the earlier claim has been corrected in the project record. The overall conclusion is that there is no statistically supported evidence of a tradable residual-reversion edge in Indian equities or single-stock futures with this signal. Every result was validated by known-answer tests, independent equity reconstruction and an adversarial audit.

Add pictures / drive links of the experimental set-up/simulations/literature review texts to support the above, if applicable

https://github.com/kartikya2207/quant-stat-arb
SURP_Kartikya

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TO BE FILLED BY THE PROFESSOR

How has the performance of the student been so far?	**Satisfactory**
Any comments/suggestions/feedback	
Signature of the professor with date	